

Insulator defect detection in transmission line in sandy and dusty scenarios based on deep learning

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ABSTRACT

Images captured by drones in complex sandy and dusty environments often have color bias and significant differences in brightness and contrast, which may make it difficult to detect insulator defects, leading to inaccurate defect detection. In this article, an enhanced defect detection algorithm relying on the mask region-based convolutional neural network (Mask R-CNN) model is put forward. The adaptive gray-world blue channel method is introduced in the image preprocessing part to improve the ability of image restoration. The VSwIn Transformer module is added to the feature extraction network. This enhances the model's ability to extract context features and identify insulator defects. Then, the remaining link between the channel attention mechanism and the spatial attention mechanism (CA-SA) is incorporated into the feature pyramid network (FPN) module in order to extract spatial information from insulators. The complete intersection over union (CIOU) loss function is utilized during the prediction phase to improve the model's convergence efficiency. Finally, the results indicate that the algorithm introduced for detecting insulator defects achieved an average accuracy of up to 89.8%, which indicates that the method can detect defective components within insulators, ensuring the safe, stable, and reliable operation of China's power grid system.

1. Introduction

Insulators are critical components of transmission lines, featuring excellent mechanical load-bearing and electrical insulation performance [1], their integrity directly impacts the operational status of the entire transmission line. Long-term exposure to the natural environment makes insulators prone to defects due to adverse weather conditions [2], especially in the areas of northwest China where sand and dust occur frequently [3,4], which may cause partial discharge, accelerate material aging, and thereby increase the risk of failure. It needs to monitor the operating status of insulators through regular inspections. Traditional insulator inspections mainly rely on manual on-site investigations, which are both ineffective and inadequate for meeting the requirements of insulator defect detection [5]. In recent years, unmanned aerial vehicle (UAV) automated inspection, which involves UAVs collecting on-site image data of overhead transmission lines and processing the image data through computer vision algorithms. This effectively addresses the challenges of manual inspections and has become one of the primary methods for line inspection. It provides a new technical means for obtaining insulator status information [6]. Therefore, accurately detecting insulator defects plays a critical role in maintaining the reliable functioning of transmission lines during UAV inspections.

The main aspects of insulator defect inspection include conventional machine learning methods and deep learning techniques [7]. Traditional machine learning methods can be categorized into three types: unsupervised learning, semi-supervised learning, and supervised learning [8–10], this method mainly relies on manual design of features and setting of specific thresholds. Zhang et al. [11] employed an improved optimal entropy thresholding (OET) method based on mathematical morphology to identify pollution levels through a deep belief network (DBN). Liu et al. [12] utilized unsupervised learning to achieve automatic defect detection on the texture surface of contact chain rod insulators. Zhang et al. [13] adopted an unsupervised image reconstruction approach to address the predicament of lacking defective data and high-quality annotations. However, most methods based on traditional machine learning are disturbed by environmental factors, resulting in reduced accuracy and robustness. Deep learning-based methods can be broadly categorized into direct two-step localization and defect localization [14,15]. Wang et al. [16] divided insulator defect detection into two stages and proposed an insulator defect detection method combining Fully Convolutional Networks (FCN) semantic segmentation and YOLOv3. Bian et al. [17] proposed a method based on Capsule Network (CapsNet) classification and YOLOv3 localization for insulator defect detection. This method can enhance the ability to find small targets, but it

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depends on the accuracy of insulator detection or segmentation. Once the results contain bounding boxes and inaccurate segmentation masks, it will have an adverse effect on the later defect diagnosis, leading to false alarms or omissions. Therefore, the two-step location defect detection methods rely on the accuracy of insulator detection, segmentation, and have a slow detection rate [18]. The direct location defect detection methods can directly obtain results at a relatively fast operation speed, this approach has become the mainstream direction of current research [19].

Direct location defect detection techniques are classified into two types: two-stage algorithms and single-stage algorithms [20]. An improved Faster R-CNN algorithm proposed by Zhao et al. [21] enhanced the detection accuracy of insulators and fault locations. Tao et al. [22] introduced an innovative multi-layer CNN cascade structure specifically for insulator positioning and flaw identification. Wan et al. [23] proposed an object detection model based on the improved YOLOv10 and introduced the MultiSEAMHead module to enhance the detection ability of occluder targets for insulator category identification and fault detection. Mahapatra et al. [24] adopted the YOLOv9 model and introduced adversarial attacks based on FGSM and PGD into the model to achieve defect classification of insulators. A model named ID-YOLO for insulation fault detection was proposed by Hao et al. [25], which addresses the issues of missed detection and false detection in complex environments. Bhagwat et al. [26] proposed a new method for detecting insulator defects in 11-kilovolt distribution systems by using the You Only Look Once (YOLOv11) and MobileNetV3 models. Now, it is possible to accurately identify insulator detection under normal weather conditions. In the face of insulator defect detection under other complex meteorological conditions, it mainly focuses on rainy, foggy, and icy weather conditions. Insulator defect detection in rainy and foggy environments is mostly carried out after image restoration. Li et al. [27] proposed attention-aligned multi-scale adversarial domain adaptation YOLO (A2MADA-YOLO), which reduces the distribution differences among foggy weather samples and enables insulator defect detection in foggy scenarios. Yang et al. [28] proposed an intelligent strategy based on a synthetic meteorological algorithm to enhance YOLOv7, enabling precise detection of insulator defects under extreme weather conditions. Ji et al. [29] proposed the DRI-Net (Derain-insulation subnet), which can effectively improve the accuracy of insulator fault detection in rainy weather. Hao et al. [30] proposed a weakly supervised transfer learning method based on YOLOv5 for identifying insulator types beneath ice and snow. Han et al. [31] utilized the Gray-Level Co-occurrence Matrix (GLCM) to determine the texture features and classification characteristics of ice layers on insulators, as well as the insulator volume extracted by traditional image segmentation algorithms.

In summary, the current research on insulator defect detection primarily focuses on normal conditions, as well as rainy and foggy weather conditions, and the algorithms adopted are generally [32] based on deep learning include RCNN and YOLO series. Despite significant breakthroughs, few studies have been conducted on insulator defect detection in sandy and dusty scenarios. Defect detection generally relies on insulator datasets, and the limited sandy and dusty insulator datasets make feature extraction difficult. In sandy and dusty images, defect detection on insulators can result in missed and false detections due to the blue channel, with large attenuation leading to a colour shift. To address the aforementioned issues, a refined Mask R-CNN algorithm is presented for insulator defect detection in sandy and dusty scenarios:

1. The adaptive gray world-blue (AGW-B) Channel algorithm is used to enhance the sandy and dusty images, correcting color deviation and contrast.
2. The VSwin Transformer module is introduced the feature extraction network to optimize the output features, allowing the model to reduce the issue of insulator leakage caused by image color deviation in sandy and dusty conditions.

3. CA-SA structure is proposed to improve the FPN module, which self-adaptively allocates the space and the channels on the attention to effectively detect small insulator targets.
4. To solve the problem of limited existing dataset, a dataset named "CPLID-Self" was constructed. The data are obtained from both image enhancement and field acquisition.

The remainder of this paper is structured as follows: Section 2 presents the improved Mask R-CNN insulator defect detection algorithm. First, we provide a summary of the Mask R-CNN framework, After further detailed analysis of the sandy images preprocessing structure of AGW-B and the BiFPNNet-based feature extraction module. Section 3 presents the complete experimental methodology, including constructing the dataset and the experimental environment, the experimental process, and the evaluation metrics. Section 4 demonstrates two aspects of the experimental results analyzed: On the one hand, by comparing with state-of-the-art image enhancement methods, it showcases the advantages of the AGW-B method in processing sandy and dusty images; On the other hand, through a comparison with cutting-edge approaches, it evaluates the algorithm's performance in insulator detection. Section 5 summarizes this full text.

2. Methods

2.1. Improved mask R-CNN algorithm

An insulator defect detection model for sandy and dusty scenes based on deep learning is shown in Fig. 1. The model adopts the improved Mask R-CNN, which consists of four parts: the image enhancement module, the feature extraction module (improved VSwin Transformer model + CA-SA model), the region proposal network (RPN [33]), and the full convolutional module. Firstly, the AGW-B [34] algorithm is used for image enhancement to preprocess the input insulator sandy and dusty image. Secondly, the VSwin Transformer model is used to obtain feature maps at different levels of insulators; the output feature maps are further fused into P2-P5 using the top-down principle. These four light gray feature maps are fused into N2-N5 using the bottom-up principle in combination with the CA-SA module. Finally, in the RPN module, predetermined anchors are set for each point in the output feature map, and multiple candidate anchors are generated afterward. Classification, bounding box regression, and high-resolution masking are used for insulator defect detection and segmentation.

2.1.1. Preprocessing of sandy and dusty images based on AGW-B

Sandy and dusty image enhancement methods can be divided into traditional methods [35–40] and data-driven methods [41–44]. Fig. 2 shows the comparison of the histogram distributions of the natural image and the sandy and dusty image.

To illustrate, in a sandy environment, an image of an insulator taken by a UAV can be represented as $x = X_{R(i,j)}, X_{G(i,j)}, X_{B(i,j)}$, where $X_{R(i,j)}$, $X_{G(i,j)}$, and $X_{B(i,j)}$ denote the pixel values of the red, green, and blue channels, respectively. Building on this representation, by analyzing the sandy and dusty images to address blue channel degradation, Our approach relies on four main insights:

1. The red channel shows superior preservation in sandy conditions compared to the blue and green channels.
2. Given blue's pronounced attenuation, we mitigate this imbalance by blending a weighted portion of the red channel into the blue channel.
3. Compensation intensity should correlate with the disparity in mean values between the blue and red channels. Specifically, under the Gray World hypothesis, this divergence directly indicates asymmetric attenuation across channels.
4. To avoid saturation during the subsequent Gray World normalization, compensation is strategically restricted to regions exhibiting significant attenuation in the blue channel.

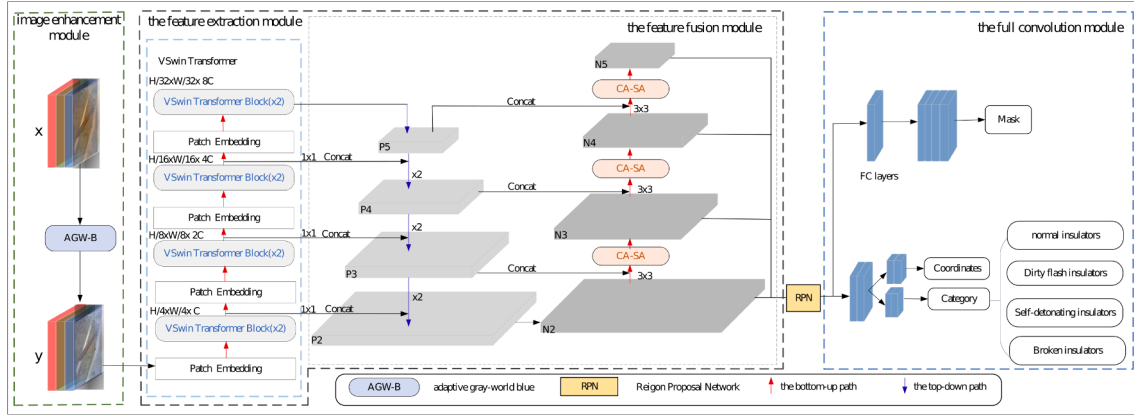


Fig. 1. Deep learning based insulator defect detection model for sandy scene.

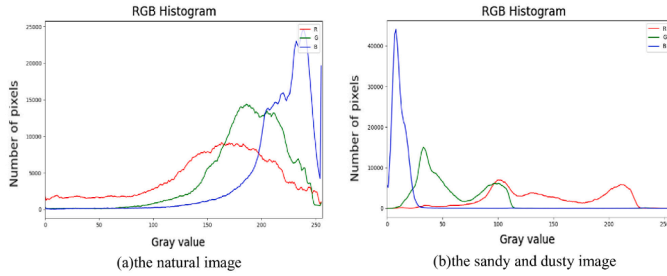


Fig. 2. Comparison of histogram distributions of the natural image and the sandy and dusty image.

Significant attenuation in the blue channel of the image is addressed by compensating this channel, with the updated blue channel value specified in Eq. (1). γ_b is a constant parameter equal to 1, demonstrating robust performance across diverse lighting conditions and acquisition setups [45]. Every component in the second term is derived directly from one of the aforementioned observations.

$$x_{B'(i,j)} = x_{B(i,j)} + \gamma_b(x_G - x_B) \cdot (1 - x_{B(i,j)}) \cdot (x_{G(i,j)}) \quad (1)$$

Where the image is divided into $M * N$ windows and the compensated local mean is calculated for each channel, L is set to 10 [46].

$$x_{(R/G/B)_{\text{Windows_mean}(i,j)}} = \frac{1}{(2L+1)^2} \sum_{(i,j) \in W_L} x_{(R/G/B)(i,j)} \quad (2)$$

Calculate the compensated global mean for each color channel.

$$x_{(R/G/B)_{\text{mean}}} = \frac{1}{MN} \sum_{i \in W} \sum_{j \in N} x_{(R/G/B')(i,j)} \quad (3)$$

Second, the Sigmoid function is used to map out the difference between the compensated local mean value and the compensated global mean value of the channel. This approach can effectively solve the existence of large local differences in luminance and contrast; and the difference is weighted to obtain the dynamic threshold difference [47]. This threshold is added to the respective compensated global mean value to determine the new compensated mean value of each pixel, thereby increasing the dynamic range width. θ is a weighting parameter, which was experimentally taken to be 0.8.

$$x_{(R/G/B)'_{\text{threshold}(i,j)}} = x_{(R/G/B)_{\text{mean}}} + (\text{Sigomid}(x_{(R/G/B)_{\text{Windows_mean}(i,j)}} - x_{(R/G/B)_{\text{mean}}}) - 0.5) \times \theta \quad (4)$$

Again, the adjustment factor m is calculated using the new compensated. By taking the values of the R, G, and B channels, which mean

value ensures that the overall brightness of the enhanced image is close to that of the original image, avoiding global over-brightness or over-darkness.

$$m = \frac{1}{3MN} \sum_{i \in W} \sum_{j \in N} [x_{R'_{\text{threshold}(i,j)}} + x_{G'_{\text{threshold}(i,j)}} + x_{B'_{\text{threshold}(i,j)}}] \quad (5)$$

Finally, the enhanced channel value is calculated $y = y_{R(i,j)}, y_{G(i,j)}, y_{B(i,j)}$.

$$y_{(R/G/B)(i,j)} = \frac{x_{(R/G/B)(i,j)}}{x_{(R/G/B)'_{\text{threshold}(i,j)}}} \cdot m \quad (6)$$

2.2. BiFPN-based feature extraction module

Using a UAV to collect insulators, as shown in Fig. 3, four different types of insulator can be obtained: Normal insulators, Dirty flash insulators, Self-detonating insulators, and Broken insulators.

Insulator defects vary considerably in size and shape. Since insulator defects (for example, dirty flash, broken) vary widely in scale (from small pixel-level cracks to large overall missing areas), the Swin Transformer can capture subtle localized defects and overall structural anomalies through a staged downsampling and windowed self-attention mechanisms. To overcome the FPN's limitation in detecting fine defects, we replace it with a BiFPN to enable better multi-scale feature fusion. In the bottom-up path, the VSwin Transformer captures long-range dependencies, improving contextual understanding of image regions. In the top-down path, the CA-SA module focuses on salient features, thereby enhancing defect localization.

2.2.1. VSwin transformer block module

The standard Swin Transformer architecture reveals obvious limitations when handling changes in the feature distribution of different defect types of insulators. Its single-window movement scheme limits the modeling ability, and cross-window connections are also rather difficult. However, the four defects of insulators often exhibit significant differences in shape, size, and texture, among other characteristics. Therefore, this paper proposes to adopt the variant Swin Transformer. Two circular shift window strategies are introduced: the Horizontal Shift Window Mechanism (HSW-MSA) and the Vertical Shift Window Mechanism (VSW-MSA). These strategies strengthen the connection between windows, break through the limitations of a single window, and expand the scope of information integration, as shown in Fig. 4. This further enhances the receptive field, helping the model obtain more abundant and comprehensive features and accurately capture the different defect features of insulators.

The conventional shift window mechanism is adopted in Fig. 5(I)(a). In Fig. 5(I)(b), the cyclic shift window technique of HSW-MSA is adopted. Regions A and C are moved to the bottom of the feature

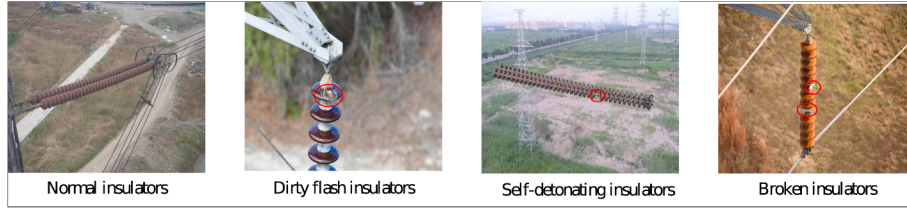


Fig. 3. Four different insulator defect categories.

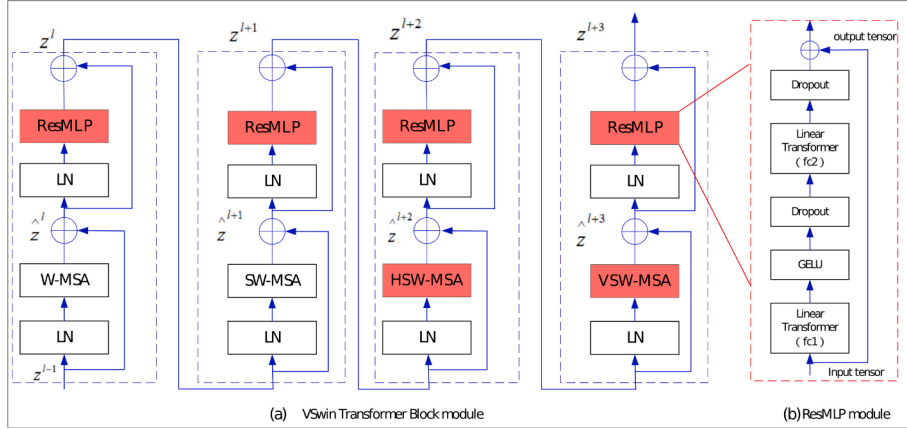


Fig. 4. Four successive VSwin transformer blocks W-MSA and SW-MSA are designed in Liang et al. (2021)[48].

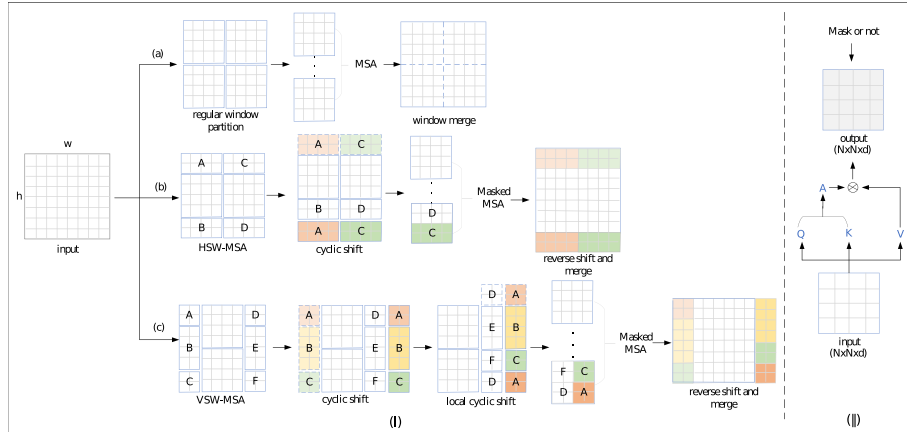


Fig. 5. Three different shift window strategies.

map and recombined with regions B and D to form a new window. In Fig. 5(I)(c), the circular shift window technique of VSW-MSA is adopted. Regions A, B, and C are moved to the right side of the feature map and recombined with regions D, E, and F to form new window. Then, D and A are moved to the lower side of the feature map and combined with F and C to form a new window. The window is adjusted using the cyclic shift technique to maintain four windows for MSA. This shift strategy connects windows and achieves global feature fusion [48], as shown in Fig. 5(II).

Finally, P_i ($i=2,3,4$) was obtained by merging the VSwin Transformer Block module [49] and summing the feature maps obtained by 2-fold upsampling of $\{P_5, P_4, P_3\}$ and the output feature map from the VSwin Transformer module. The generation process for each feature layer in the top-down path is described in the following Eqs. (7) ~ (8):

$$P_5 = VST \quad (7)$$

$$P_i = F_{1 \times 1}(VST) \oplus U_2[P(i+1)] \quad (8)$$

Where $F_{1 \times 1}$ denotes 1×1 convolution. The symbol $\oplus$ denotes the addition of elements [50], $U_j(x)$ denotes that the feature mapping x is upsampled by a factor j .

2.2.2. Feature fusion module introduced by CA-SA

Attention mechanism [51] (AM) can be split into three types: Channel Attention (CA), Spatial Attention (SA), and mixed attention. Insulator edges are prone to peeling, chipping, and breakage, resulting in small target defects that are difficult to detect. By borrowing the network architecture from PANet, a CA-SA module embedding a combination of channel attention and spatial attention is added to the right side of the BiFPN module, as shown in Fig. 1. The CA-SA module represents an input feature passing through the parallel residual module of CAM and SAM to obtain an output feature map, as shown in Fig. 6. N_j ($j=3,4,5$) are used to denote the mapping of each layer in the bottom-up path of the feature; this process corresponds to:

$$N_2 = P_2 \quad (9)$$

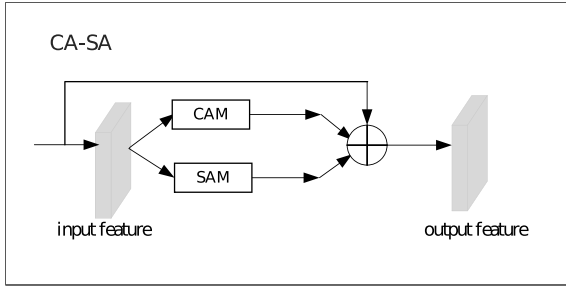


Fig. 6. CA-SA module.

$$Nj = W_c[F_{3 \times 3}[N](j-1)] + W_s[F_{3 \times 3}[N](j-1)] + F_{3 \times 3}[N](j-1) \oplus Pj \quad (10)$$

Where $F_{3 \times 3}$ denotes 3×3 convolution. $W_c(x)$ denotes the features obtained by feature x through the CAM module, and $W_s(x)$ denotes the features obtained by feature x through the SAM module. Finally, four different scales of feature mappings $y' = N2, N3, N4, N5$ are used for fault detection.

3. Experimental results

3.1. Dataset and experimental environment

The experiments were performed on a Windows 11 platform. For training and evaluation, a laptop with an NVIDIA GeForce RTX 4090 GPU and a 12th Gen Intel Core i7-12700K/KF processor was used. The setup included CUDA 12.1 and cuDNN 8.9.7, with implementations in Python 3.9 and the PyTorch 2.4.1 framework.

Existing insulator defect datasets primarily cover general weather conditions, with few examples of weather conditions involving sandy or dusty. To address this, we augmented images from the CPLID [52] and IDID [53] datasets using Imgaug and added real sandy weather images collected separately. The resulting dataset combines these sources in a 7:3 ratio of simulated-to-real sandy weather images. The CPLID and IDID datasets have a total of 1027 images. A DJI M30T UAV device was used to collect the self-acquisition dataset, as shown in Fig. 7(a), which includes 349 dusty drone images. This article merges datasets containing two types of insulators, referred to as the “CPLID-Self”, which include 899 normal insulator aerial images and 477 insulator defect aerial images. Through vertical and horizontal flipping, the “CPLID-Self” dataset was further expanded, as shown in Fig. 7(b), resulting in a total of 2557 images. Among them, the dataset is split into training, validation, and test sets in an 8:1:1 ratio, with 2045, 255, and 257 images, respectively. All images in the “CPLID-Self” dataset are 224×224 pixels.

3.2. Test procedure

1) Labeling of insulators

Labeling software was chosen to add sample annotations to the photos in the insulator dataset. All the labeling information will be saved to the json file corresponding to the original image. Labeling annotation of 0051.jpg in the sandy insulators image dataset is exemplified in Fig. 7(c).

2) Image pre-processing

An original image of a sandy insulator is input, and the AGW-B algorithm is used to clarify the input image to get a normal color insulator picture, as shown in Fig. 8(II).

3) Improved Mask R-CNN model training

The training set data were input into the deep learning-based insulator defect detection model for sandy and dusty scenes for training. The VSwin Transformer backbone underwent 300-epoch pretraining on ImageNet-1K (Deng et al., 2009 [54]). Subsequently, the Cas-VSwin architecture was fine-tuned on COCO (Lin et al., 2014 [55]) using the

Table 1

Results of instance segmentation on the surface defect validation set by different initial weights.

Initialization	Box AP	Mask AP
VSwin(ImageNet-1 K)	86.3	82.9
VSwin(COCO)	85.2	80.8
VSwin(Random)	83.9	80.7

MMDetection (Chen et al., 2019 [56]) framework under multi-scale training protocol. Table 1 reveals that models pretrained on ImageNet achieve significant gains of +2.4 Box AP and +2.2 Mask AP compared to random initialization, along with improvements of +1.1 Box AP and +2.1 Mask AP over COCO pretraining. Analyzing the detection results, the trained convolutional neural model has a positive impact on insulator defect detection, with a low misjudgment rate, and it can effectively recognize different kinds of insulator defects, as shown in Fig. 8(III). The trained model is verified with the validation set data for model overfitting and then tested with the test data set. As shown in Fig. 9, after 40 iterations, the total loss curves of the training and validation sets are similar and decrease to less than 0.3, oscillating within a small range. This indicates that the network produces correct outputs for most image samples, and the model parameters do not overfit during training, demonstrating a strong ability to recognize insulator defects.

3.3. Evaluation indicators

Evaluation metrics as a quantitative evaluation to assess the detection performance of insulator defects in sandy scenarios are divided into two categories. One class is the quantitative evaluation of dusty image enhancement. Four metrics were utilized for quantitative assessment: the Peak Signal-to-Noise Ratio (PSNR) [57], the Structure Similarity Index Measure (SSIM) [58], the Multi-Scale Structural Similarity Index Measure (MS-SSIM) [59], and the Chromatic Difference Indicator (CIEDE) [60]. PSNR, SSIM, and MS-SSIM are defined as Eqs. (11) ~ (12) ~ (13). Better enhancement performance can be indicated by higher PSNR and SSIM values, and higher image quality can be indicated by lower CIEDE score values.

$$PSNR = 20 \cdot \log_{10} \left(\frac{MAX_I}{MSE} \right) \quad (11)$$

$$SSIM(x, y) = [l(x, y)]^\alpha \cdot [c(x, y)]^\beta \cdot [s(x, y)]^\gamma \quad (12)$$

$$MS-SSIM = [l_M(x, y)]^{\alpha_M} \prod_{j=1}^M [c_j(x, y)]^{\beta_j} [s_j(x, y)]^{\gamma_j} \quad (13)$$

where MAX_I represents the maximum achievable pixel value in the image, and MSE is the mean squared error, representing the average squared difference in pixel values between two images. Where $l(x, y)$, $c(x, y)$, and $s(x, y)$ represent three independent components extracted at a single spatial scale (resolution): luminance, contrast, and texture; the texture similarity in IQA proposed by Wang et al. [58] can be set as $\alpha = \beta = \gamma = 1$. Where c_j and s_j denote the contrast and structural similarity calculated at each scale j , respectively, while brightness similarity is computed solely at the proportionality scale M by $l_M(x, y)$. The exponents α_M , β_j , and γ_j adjust the relative importance of different components. The parameter $\alpha_M = \beta_j = \gamma_j, j = 5$ is set as in Wang et al. [61]. Subsequently, the overall cross-scale evaluation is derived through the weighted product of the aforementioned components extracted at different scales.

The other class is the quantitative evaluation of insulator defect detection. Precision (P), Recall (R), Average Precision (AP), Mean Average Precision (mAP0.5), and the harmonic mean (F1-score, F1) were adopted for quantitative evaluation.

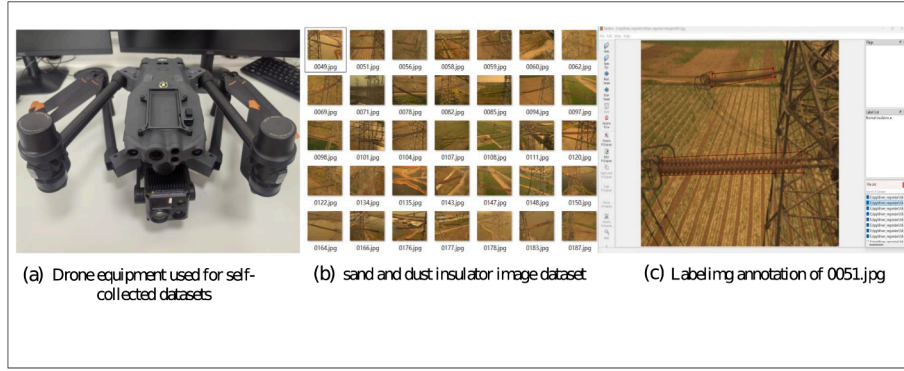


Fig. 7. Example of part of the dataset and labeling of insulator defect samples.

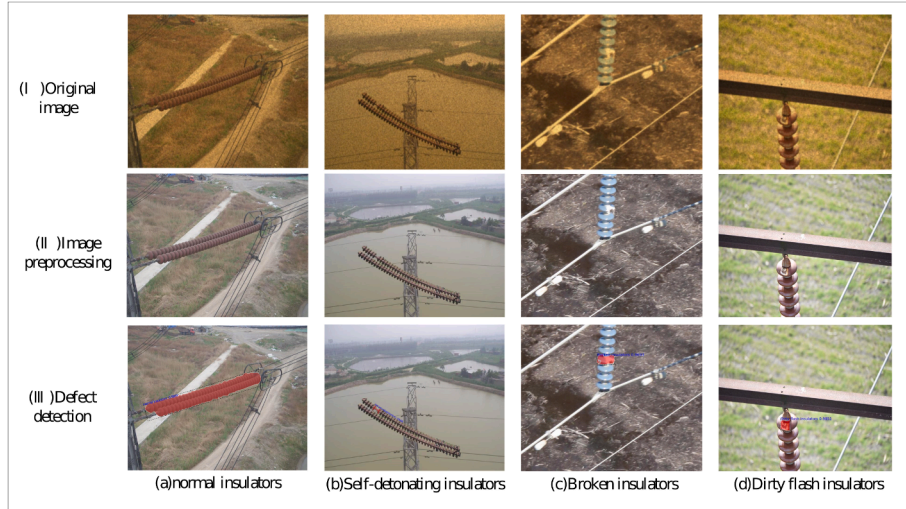


Fig. 8. Inspection results of insulators under different defects.

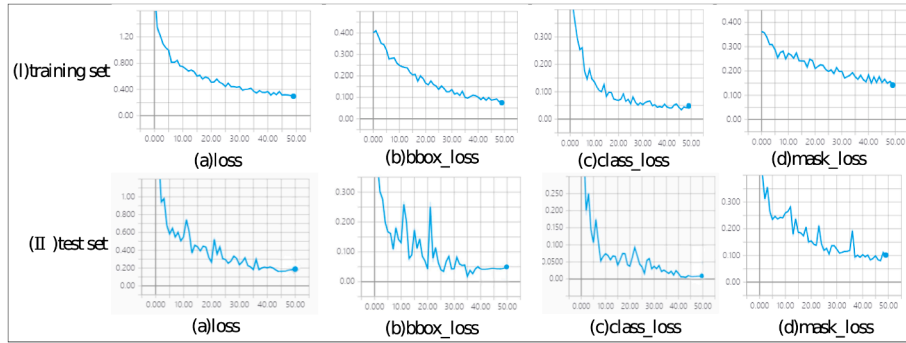


Fig. 9. Loss vs. number of iterations for the improved Mask r-cnn.

4. Evaluation of experimental outcomes

4.1. Comparison with the latest dusty image enhancement methods

The proposed AGW-B method is compared with both traditional and data-driven state-of-the-art image enhancement algorithms. This comparison utilizes quantitatively and qualitatively synthesized sandy and dusty images to demonstrate the effectiveness of the proposed method in enhancing such images. The competing algorithms include traditional-based algorithms (Park and Eom[39], Shi et al. [40]) and data-driven-based approaches (Meng et al. [41], Guo et al. [43]).

The quantitative evaluation results are given in Table 2. AGW-B method is compared with the objective metrics of the existing state-

of-the-art sandy and dusty image enhancement methods. It can be observed that the method of this paper outperforms all the given methods in both PSNR and CIEDE metrics. Although slightly lower than Guo et al. [43] method in terms of SSIM, our approach achieves optimal performance on the MS-SSIM metric, highlighting the superiority of MS-SSIM as an evaluation standard. This metrics indicates that the proposed architecture can effectively remove the color bias, reduce the large local differences in picture brightness and contrast, and achieve good enhancement results. The qualitative evaluation results are given in Fig. 10, which shows that the methods based on traditional algorithms can hardly achieve image dust removal. These methods only reduce the dusty color and fail to completely remove the yellow color bias in sandy and dusty images, such as Park and Eom[39], Shi et al. [40]. The meth-

Table 2
Quantitative evaluation of enhanced dusty images.

Evaluation	This paper	Park and Eom [39]	Shi et al. [40]	Meng et al. [41]	Guo et al. [43]
PSNR↑	23.1235	14.5155	14.9339	19.51	22.1383
SSIM↑	0.8436	0.7586	0.7392	0.8086	0.8544
MS-SSIM↑	0.8785	0.8012	0.7568	0.8148	0.8649
CIEDE↓	10.31	-	-	11.22	-

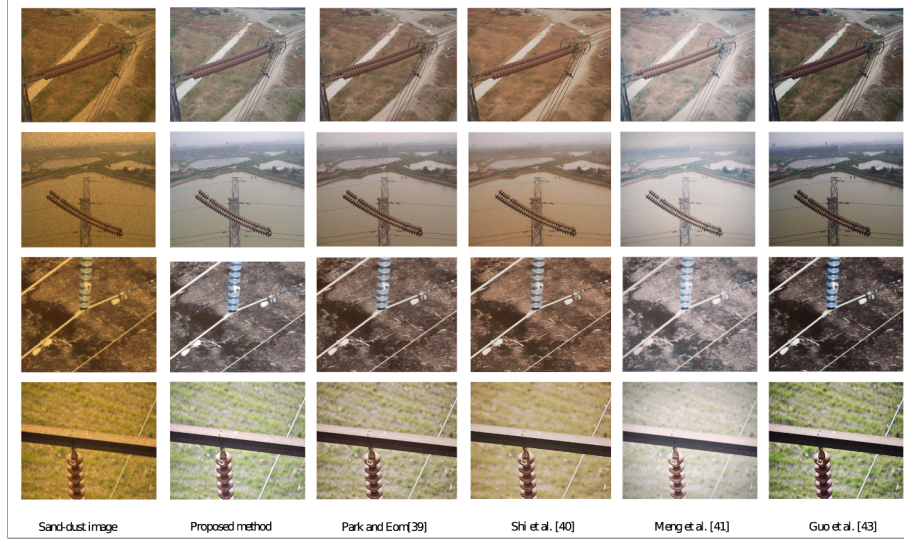


Fig. 10. dusty image enhancement results of different methods.

ods of Meng et al. [41] and Guo et al. [43], and the algorithms proposed in this paper can completely remove the dusty color, but Meng et al. [41] makes the image still have a lot of smoke residue, and the method of Guo et al. [43] has a lower image brightness and a larger contrast difference.

4.2. Comparison with the latest insulator defect detection methods

1) Performance Comparison Experiment

Experiments are conducted to evaluate insulator detection performance by comparing the proposed model with current advanced insulator defect detection approaches. The comparison includes two-stage algorithms (Zhao et al. [21], Tao et al. [22]) and a single-stage algorithms (Wan et al. [23], Mahapatra et al. [24], Hao et al. [25], Bhagwat et al. [26], and Jocher et al. [62]). All models are trained using the data enhancement strategy of the proposed AGW-B module. The seven compared methods use similar training and testing backgrounds.

From the data in Table 3 can be clearly seen in the insulator defect detection tasks, the proposed method demonstrates significant advantages across four core metrics: Precision, Recall, mAP0.5, and F1-score. For normal insulators, the proposed method achieves a precision rate that increases from 76.1% in Jocher et al. [62] from 76.1% to 96.7%, recall from 77.0% to 95.3%, mAP0.5 from 78.3% to 98.8%, and F1-score from 0.765 to 0.975. This achieves precise identification and comprehensive coverage of the target. For low-distinctiveness dirty flash insulators, the precision rate improved from 66.8% of Tao et al. [22] to 90.8%, while the recall rate increased from 51.1% to 87.2%. mAP0.5 increased from 56.7% to 89.4%, and the F1-score improved from 0.583 to 0.901. This effectively resolves the issues of missed detections and false positives caused by the blurred appearance of flashover defects. Compared to the latest YOLOv10 approach by Wan et al. [23], the proposed method achieves comprehensive optimization across core metrics; For normal insulators, it not only significantly outperforms in all metrics but also improves the F1-score from YOLOv10's 0.949 to 0.975; For

dirty flash insulators, metrics show steady improvement, with an F1-score of 0.901 achieving an effective gain over its 0.891 baseline; For self-detonating defects, metrics are substantially surpassed, with the F1-score rising from 0.917 to 0.936, an improvement of 0.019, accurately capturing local defect features. Compared to Mahapatra et al. [24], the proposed method demonstrates superior completeness and robustness. A key advantage is its ability to detect all defect categories, whereas their approach fails to report results for self-detonating insulators. Compared to Bhagwat et al. [26], the proposed method shows clear advancements, particularly in handling environmentally degraded cases. Notably, it provides complete results for dirty flash insulators, where its approach provides no reported results. For broken insulators, metrics show slight enhancement, with the F1-score increasing from 0.785 to 0.809, an increase of 0.024, strengthening fine-grained recognition capabilities for damaged areas. Additionally, for broken insulators, the proposed method achieved a significant improvement in accuracy, increasing from 59.1% of Tao et al. [22] to 78.7%, recall from 39.9% to 75.3%, mAP0.5 from 41.7% to 76.3%, and F1-score from 0.476 to 0.809, enabling the detection of minor defects in insulators. The results show that the method proposed in this paper improves the recognition rate and effectively enhances the recognition of insulators in sandy and dusty scenarios, especially for the difficult to recognize flashover damage and insulator breakage defects. It achieves a higher recognition rate and demonstrates practical application value.

Comparative target detection experiments were conducted for Meng et al. [41] and Guo et al. [43] under sandy and dusty bad weather conditions to verify the effect of image restoration on the subsequent target detection algorithms. The results of the different dusty removal models are displayed in Fig. 11, illustrating that the improved Mask R-CNN algorithm not only detects and processes weather more efficiently, but also achieves higher accuracy and lower leakage rate compared to other algorithms under sandy and dusty weather conditions.

2) ablation experiment

Table 3
Comparative experiments on the performance of insulators.

Model	Class	Precision (%)	Recall (%)	mAP _{0.5} /%	F1-score
Zhao et al. [21]	normal insulators	87.0	89.5	85.6	0.826
	Dirty flash insulators	72.0	63.9	65.4	0.677
	Self-detonating insulators	82.1	74.7	81.0	0.876
	Broken insulators	71.5	60.1	61.2	0.650
Tao et al. [22]	normal insulators	90.1	93.4	93.7	0.917
	Dirty flash insulators	66.8	51.1	56.7	0.583
	Self-detonating insulators	98.3	98.8	99.5	0.969
	Broken insulators	59.1	39.9	41.7	0.476
Wan et al. [23]	normal insulators	93.5	94.2	95.1	0.949
	Dirty flash insulators	88.6	85.4	86.9	0.891
	Self-detonating insulators	80.2	88.7	85.7	0.917
	Broken insulators	77.3	74.6	75.1	0.785
Mahapatra et al. [24]	normal insulators	94.1	95.2	97.8	0.974
	Dirty flash insulators	89.8	78.8	81.7	0.859
	Self-detonating insulators	-	-	-	-
	Broken insulators	78.6	75.1	75.9	0.799
Hao et al. [25]	normal insulators	73.0	75.6	76.2	0.742
	Dirty flash insulators	72.3	72.1	72.0	0.721
	Self-detonating insulators	76.9	65.5	74.8	0.707
	Broken insulators	70.3	69.1	70.9	0.696
Bhagwat et al. [26]	normal insulators	95.9	95.3	98.5	0.973
	Dirty flash insulators	-	-	-	-
	Self-detonating insulators	95.9	89.5	93.7	0.907
	Broken insulators	75.8	74.9	75.6	0.781
Jocher et al. [62]	normal insulators	76.1	77.0	78.3	0.765
	Dirty flash insulators	74.0	74.5	79.9	0.778
	Self-detonating insulators	77.5	70.6	75.1	0.738
	Broken insulators	72.7	71.3	73.4	0.691
Proposed method	normal insulators	96.7	95.3	98.8	0.975
	Dirty flash insulators	90.8	87.2	89.4	0.901
	Self-detonating insulators	96.9	90.5	94.9	0.936
	Broken insulators	78.7	75.3	76.3	0.809

Table 4
Sandy and dusty insulator ablation experiments comparison.

Baseline Model	AGW-B	Back-bone	CA-SA	Loss	NI (%)	DFI (%)	SDI (%)	BI (%)	all (%)
MaskR-CNN		Resnet50		Smooth L1	93.9	82.9	90.1	70.4	85.5
	✓	Resnet50		Smooth L1	94.5	85.1	93.9	71.1	86.1
	✓	ST		Smooth L1	93.9	85.2	94.1	72.9	86.2
	✓	ST + HSW		Smooth L1	94.2	85.6	94.5	74.4	87.1
	✓	ST + VSW		Smooth L1	94.4	85.7	94.7	74.6	87.3
	✓	VST		Smooth L1	94.6	85.9	94.8	75.6	87.7
	✓	VST	✓	Smooth L1	97.5	87.9	95.9	75.9	88.1
	✓	VST	✓	GIoU	97.9	88.5	94.9	76.2	88.6
	✓	VST	✓	CloU	98.8	89.4	94.9	76.3	89.4

A series of ablation experiments were conducted to assess the performance of the enhanced module. The results were compared to those of the baseline network in identifying insulator defects under sandy and dusty conditions. The findings from these experiments are presented in Table 4. For the ablation study, the original Mask R-CNN algorithm utilizing ResNet50 as its backbone network. Nine distinct sets of ablation algorithms were established for this purpose.

Note: “✓” denotes the model adding module, NI denotes “normal insulator”, DFI denotes “Dirty flash insulators”, SDI denotes “Self-detonating insulators”, BI denotes “Broken insulators”, ST denotes “Swin Transformer”, and VST denotes “VSwIn Transformer”.

From Table 4, the average detection accuracy of the baseline model (Mask R-CNN + Resnet50 + Smooth L1 loss) is 85.5%. After simply introducing the AGW-B module, the average detection accuracy is improved to 86.1%, which is 0.6 percentage points higher than the previous value. This is because after the introduction of the AGW-B module,

the model can effectively eliminate the dust interference in the dust environment, solve the problems of target deviation, blur and low contrast, and lay a foundation for further identification of defects in insulators.

Based on the introduction of the AGW-B module, by gradually replacing the backbone network from Resnet50 to the standard Swin Transformer (ST), ST + HSW, ST + VSW, and finally adopting VSwIn Transformer (VST). The average detection accuracy of the improved Mask R-CNN network has further increased to 87.7%, which is 1.6 percentage points higher than 86.1% of AGW-B + Resnet50. Compared with 86.2% when only the standard Swin Transformer (ST) is used, it is further 1.5 percentage points higher. This is mainly attributed to the introduction of HSW and VSW modules by VSwIn Transformer on the basis of standard Swin Transformer, effectively compensating for the deficiency of standard Swin Transformer in capturing insulator defect features. It has significantly enhanced the efficiency of identifying and distinguishing various defect features in complex sand and dust environments, thereby achieving a continuous improvement in detection performance.

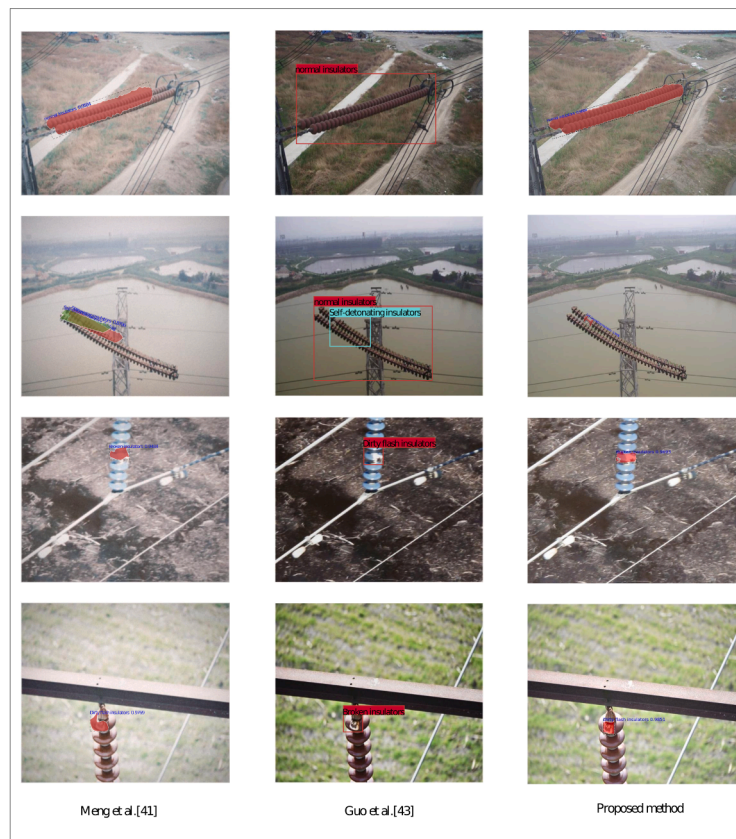


Fig. 11. Visual comparison of detection results using Meng et al. [41], Guo et al. [43], Proposed method.

Subsequently, by adding the self-attention mechanism CA-SA module to the model, the average detection accuracy rate increased from 87.7% to 88.1%, indicating that the CA-SA module is more effective than the traditional channel attention mechanism. The loss functions were replaced from Smooth L1 to GIoU and CIoU. The experimental results showed that the result of using the CIoU loss function reached 89.4%, which was 1.3% higher than the 88.1% of Smooth L1 and 0.8% higher than the 88.6% of the GIoU loss function. This indicates that this loss function can effectively improve the positioning accuracy of insulator defect detection, verifying the rationality of the selection of this loss function.

To sum up, the results of the ablation test show that the proposed improved Mask R-CNN model plays a role in all three improvements, and although there is a reduction in efficiency, the detection accuracy is significantly improved, and it is more effective for the detection of different types of insulators.

5. Conclusion

Aiming at the problems of image quality degradation of insulators and difficulty in detecting small target defects in sandy and dusty environments, this paper proposes an insulator defect detection model based on the improved Mask R-CNN. The main innovations and contributions of this method are as follows:

1. The AGW-B module was introduced to enhance the network's feature adaptability under the interference of sand and dust, effectively alleviating the problems of missed detections and false detections.
2. The VSwin Transformer module is adopted to replace the Resnet50 convolutional structure, enhancing the extraction accuracy of long-distance semantic dependencies and small target features.

3. By integrating the CA-SA multi-attention mechanism in PANet and constructing an adaptive feature pyramid, the focusing and fusion capabilities for multi-scale insulator defects have been enhanced.
4. Comprehensive verification was conducted on public datasets (CPLID, IDID) and self-built sand and dust environment datasets. This method outperformed mainstream baseline models in multiple evaluation metrics.

The experimental results show that the model proposed in this paper has a significant performance improvement in the sandy and dusty environment: the AGW-B module increases the average detection accuracy of insulator defects by 0.6%; the backbone network of VSwin Transformer has increased the average detection accuracy from 86.1% to 87.7%; after combining the CA-SA mechanism with the GIoU loss function, the overall accuracy of the model reached 88.6%; the overall accuracy of the final complete model on public datasets (CPLID, IDID) and self-built sand and dust datasets reached 89.4%, which was 3.9% higher than the comprehensive performance of the original Mask R-CNN. Therefore, the proposed model has stronger feature adaptability and detection robustness in complex sand and dust environments, and can provide reliable technical support for identifying insulation defects in electrical testing systems.

In the subsequent research, we will consider defect detection of insulators under other extreme weather (rainy, foggy, snowy and icy), and try to construct a suitable transfer learning network, which makes it possible to detect the defect detection of insulators under different extreme weather.

CRediT authorship contribution statement

Xiaoping Li: Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Con-

ceptualization; **Yujie Sun**: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization; **Xinyue Liu**: Writing – review & editing, Supervision, Project administration; **Xun Wu**: Writing – review & editing, Supervision, Project administration.

Data availability

The data has been explained in the article.

Declaration of competing interest

We declare the following interests related to this submission: No financial interests/personal relationships to declare. All data used in this study are original and generated through simulations in public repositories.

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